

Chapter # 14

Exercise 14.2

Evaluate the iterated integral.

$$3: \int_0^3 \int_0^{\sqrt{9-y^2}} y \, dx \, dy = \int_0^3 y \left[x \right]_0^{\sqrt{9-y^2}} dy$$

$$= \int_0^3 y \left[\sqrt{9-y^2} - 0 \right] dy$$

$$= \int_0^3 y \sqrt{9-y^2} \, dy = \int_0^3 (9-y^2)^{1/2} (-2y) \, dy$$

$$= -\frac{1}{2} \left[\frac{(9-y^2)^{1/2+1}}{\frac{1}{2}+1} \right]_0^3$$

$$= -\frac{1}{2} \times \frac{2}{3} \left[(9-y^2)^{3/2} \right]_0^3 = -\frac{1}{3} \left[(9-(3)^2)^{3/2} \right]$$

$$\left(9-(0)^2 \right)^{3/2}$$

$$= -\frac{1}{3} \left[0 - (5)^{3/2} \right] = \frac{-1 \times -27}{3} = 9$$

$$= 9$$

Double integrals over non-rectangular Regions

$$5: \int_{\sqrt{\pi}}^{\sqrt{2\pi}} \int_0^{x^3} \sin(y/x) dy dx = \int_{\sqrt{\pi}}^{\sqrt{2\pi}}$$

$$= \int_{\sqrt{\pi}}^{\sqrt{2\pi}} \left[-\cos(y/x) \cdot \frac{1}{x} \right]_0^{x^3} dx \quad \because \sin x = -\cos x$$

$$\sin x^2 = -\frac{\cos x^2}{2x}$$

$$= \int_{\sqrt{\pi}}^{\sqrt{2\pi}} -x \left[\cos(y/x) \right]_0^{x^3} dx$$

$$= \int_{\sqrt{\pi}}^{\sqrt{2\pi}} -x \left[\cos\left(\frac{x^3}{x}\right) - \cos\left(\frac{0}{x}\right) \right] dx$$

$$\text{let } u = x^2, \quad du = 2x dx = \frac{1}{2} du = dx$$

$$\int_{\sqrt{\pi}}^{\sqrt{2\pi}} \left[-x \cos(x^2) + x \right] dx \quad \int -x \cos x^2 dx = \int -u \cos u \cdot \frac{1}{2} du$$

$$\int \cos u du = -\frac{1}{2} \sin u$$

$$= \left[-\frac{1}{2} \sin(x^2) + \frac{x^2}{2} \right]_{\sqrt{\pi}}^{\sqrt{2\pi}} = -\frac{1}{2} \sin x^2$$

$$= \left[-\frac{1}{2} \sin(\sqrt{2\pi})^2 + \frac{(\sqrt{2\pi})^2}{2} \right] - \left[-\frac{1}{2} \sin(\sqrt{\pi})^2 + \frac{(\sqrt{\pi})^2}{2} \right]$$

$$= (0 + \pi) - (-0 + \frac{\pi}{2}) = \pi - \frac{\pi}{2} = \frac{\pi}{2}$$

$$6. \int_{-1}^1 \int_{-x^2}^{x^2} (x^2 - y) dy dx = \int_{-1}^1 \left[\int_{-x^2}^{x^2} (x^2 - y) dy \right] dx$$

$$= \int_{-1}^1 \left[\int_{-x^2}^{x^2} (x^2 - y) dy \right] dx$$

$$= \int_{-1}^1 \left[x^2 y - \frac{y^2}{2} \right]_{-x^2}^{x^2} dx$$

$$= \int_{-1}^1 \left[\left(x^2(x^2) - \frac{(x^2)^2}{2} \right) - \left(x^2(-x^2) - \frac{(-x^2)^2}{2} \right) \right] dx$$

$$= \int_{-1}^1 2x^4 dx$$

$$= \left[2x \frac{5}{5} \right]_{-1}^1 = \frac{2}{5} \left[(1)^5 - (-1)^5 \right]$$

$$= \frac{2}{5} [1 + 1]$$

$$= \frac{4}{5}$$

$$8: \int_1^2 \int_0^{y^2} e^{x/y^2} dx dy$$

$$\int_1^2 \left[y^2 \int_0^{y^2} e^{x/y^2} \cdot \frac{1}{y^2} dx \right] = \int_1^2 y^2 \left[e^{x/y^2} \right]_0^{y^2}$$

$$= \int_1^2 \left[y^2 \left[e^{y^3/y^2} - e^{0/y^2} \right] dy \right]$$

$$= \int_1^2 \left[y^2 \left[e^y - e^0 \right] dy \right]$$

$$= \int_1^2 (e-1) y^2 dy = (e-1) \left[\frac{y^3}{3} \right]_1^2$$

$$= \frac{(e-1)}{3} \left[(2)^3 - (1)^3 \right]$$

$$\frac{7(e-1)}{3}$$

Exercise 14.2

$$7: \int_0^1 \int_0^x y \sqrt{x^2 - y^2} dy dx$$

$$= \int_0^1 \left[\int_0^x y \sqrt{x^2 - y^2} dy \right] dx$$

$$= \int_0^1 \left[-\frac{1}{2} \times \frac{2}{3} (x^2 - y^2)^{3/2} \right] dx$$

$$= \int_0^1 \left[-\frac{1}{3} \left[(x^2 - (x)^2)^{3/2} - (x^2 - (0)^2)^{3/2} \right] \right] dx$$

$$= \int_0^1 \frac{1}{3} x^3 dx = \frac{1}{3} \left[\frac{x^4}{4} \right]_0^1 = \left[\frac{x^4}{12} \right]_0^1$$

$$= \frac{1}{12} \left[(1)^4 - (0)^4 \right]$$

$$= \frac{1}{12}$$

19: $\int_R x(1+y^2)^{-1/2} dA$; R is the region in

the first quadrant enclosed by $y=x^2$, $y=4$ and $x=0$.

$$\int_0^4 \int_0^{\sqrt{y}} x(1+y^2)^{-1/2} dx dy$$

$$= \int_0^4 \frac{1}{2} y (1+y^2)^{-1/2} dy$$

$$= \frac{\sqrt{17} - 1}{2}$$

$$56: \iint_D x \, dA = \int_1^3 \int_0^{\ln y} x \, dy \, dx$$

$$\int \int_D x \, dA = \int_0^{\ln 3} \int_{e^y}^3 x \, dx \, dy$$

$$= \int_0^{\ln 3} \left[\frac{x^2}{2} \right]_{x=e^y}^{x=3} dy$$

$$= \frac{1}{2} \int_0^{\ln 3} 9 - e^{2y} dy$$

$$= \frac{1}{2} \left[-\frac{1}{2} e^{2y} + 9y \right]_0^{\ln 3}$$

$$= \frac{1}{2} \left[-\frac{1}{2} e^{2\ln 3} + 9\ln 3 \right] - \frac{1}{2} \left[0 - \frac{1}{2} \right]$$

$$= \frac{1}{2} \left[9\ln 3 - \frac{9}{2} + \frac{1}{2} \right]$$

$$= \boxed{-2 + \frac{9}{2} \ln 3} \quad \text{Ans}$$